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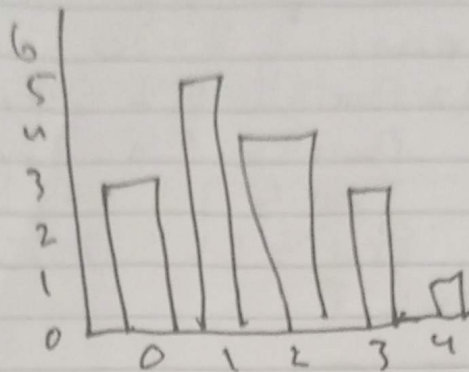
No

Date

PCD

1. latihan soal: barisan 256 x 256 piksel
 dengan ukuran 4x4 dan memiliki 5 gray scale
 dimana 120 citra akan disegmentasikan
 menggunakan metode thresholding

0	1	1	0
1	3	4	3
2	2	3	2
1	0	2	1



- a. tentukan nilai thresholding dari blok citra (u) dimana citra asli dibagi menjadi 2 blok citra secara vertikal kemudian tulis hasil citra setelah thresholding.

• Matrix Block (B_i)

↳ menghitung nilai thresholding (T_i)

~~$$T_i = \frac{0+1+1+3+2+2+1+0}{8} = \frac{10}{8} = 1.25$$~~

$$T_i = \frac{0+1+1+3+2+2+1+0}{8} = \frac{10}{8} = 1.25$$

$$B_1 =$$

0	1
1	3
2	2
1	0

Thresholding pada blok 1

nilai pixel $\geq 1.25 \rightarrow 1$

nilai pixel $< 1.25 \rightarrow 0$

maka :
 = Baris 1 : 0 $\rightarrow$ 0, 1 $\rightarrow$ 0
 = Baris 2 : 1 $\rightarrow$ 0, 3 $\rightarrow$ 1
 = Baris 3 : 2 $\rightarrow$ 1, 2 $\rightarrow$ 1
 = Baris 4 : 1 $\rightarrow$ 0, 0 $\rightarrow$ 0

Hasil

0	0
0	1
1	1
0	0

matrik B_{32} 2 (B_2)

nilai thresholding (T_2) $B_2 =$

$$T_2 = \frac{1+0+4+3+3+2+2+1}{8}$$

$$= \frac{16}{8} = 2$$

1	0
4	3
3	2
2	1

Thresholding Berdasarkan 2

nilai pixel $\geq 2 \rightarrow 1$

nilai pixel $< 2 \rightarrow 0$

maka = Baris 1 = 1 $\rightarrow 0$, 0 $\rightarrow 0$

Baris 2 = 4 $\rightarrow 1$, 3 $\rightarrow 1$

Baris 3 : 3 $\rightarrow 1$, 2 $\rightarrow 1$

Baris 4 : 2 $\rightarrow 1$, 1 $\rightarrow 0$

Hasil akhir:

0	0	0	0
0	1	1	1
1	1	1	1
0	0	1	0

6. Jika diketahui within-class variance $T(0) = 1,36$
 $T(1) = 0,87$, $T(2) = 0,76$ dan $T(4) = 1,16$, maka
 tentukan within class variance $T(2)$ dengan
 menggunakan Thresholding

• Background

pixel dengan level 1 (grey-level 0 level 1, dan)

total pixel background (n_b) = 3 + 5 + 4 = 12 pixel

menghitung bobot (weight, w_b)

$$w_b = \frac{n_b}{n} = \frac{12}{16} = 0,75$$

menghitung rata-rata

$$M_b = \frac{(0.3) + (1.5) + (2.4)}{12}$$

$$M_b = \frac{0 + 5 + 0}{12} = \frac{13}{12} = 1.0833$$

Menghitung variansi (varianci, σ^2_b)

$$\sigma^2_b = \frac{(11 - 1.0833)^2 \cdot 3) + ((11 - 1.0833)^2 \cdot 5) + ((12 - 1.0833)^2 \cdot 4)}{12}$$

$$\sigma^2_b = \frac{6.9162}{12} \approx 0.5764$$

• Foreground

Piksel dengan level ≥ 3 (yaitu level 3 dan 4)Total piksel Foreground (n_F) = $3 + 1 = 4$ pikselmenghitung bobot (weight, w_F)

$$w_F = \frac{n_F}{n} = \frac{4}{16} = 0.25$$

menghitung rata-rata (m_F , m_F)

$$m_F = \frac{(3 \cdot 3) + (4 \cdot 1)}{4} = \frac{9 + 4}{4} = \frac{13}{4} = 3.25$$

menghitung variansi (varianci, σ^2_F)

$$\sigma^2_F = \frac{((3 - 3.25)^2 \cdot 3) + ((4 - 3.25)^2 \cdot 1)}{4}$$

$$\sigma^2_F = \frac{0.75}{4} = 0.1875$$

Perhitungan akhir within-class variance $\sigma^2_w(3)$

$$\sigma^2_w = (0.75 \cdot 0.15764) + (0.25 \cdot 0.1875)$$

$$\sigma^2_w = 0.14792$$

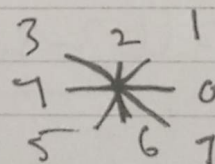
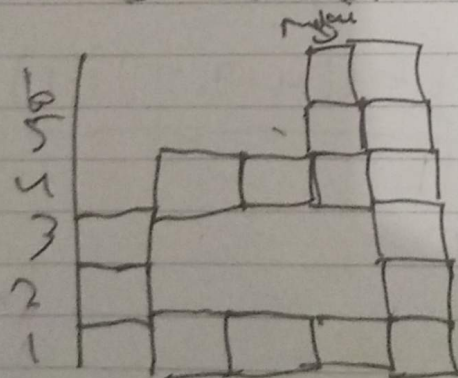
c. berdasarkan hasil pada soal no. b tentukan nilai thresholding terbaik menggunakan Otsu thresholding dan gambarkan citra setelah thresholding

Baris 1: $1 < 2-70, 1 < 2-70, 1 < 2-70, 0 < 2-70$
 Baris 2: $1 < 2-70, 3 \geq 2-74, 4 \geq 2-71, 3 \geq 2-74$
 Baris 3: $2 \geq 2-71, 2 \geq 2-71, 3 \geq 2-71, 2 \geq 2-71$
 Baris 4: $1 < 2-70, 0 < 2-70, 2 \geq 2-71, 1 < 2-70$

Citra hasil

0	0	0	0
0	1	1	1
1	1	1	1
0	0	1	0

2. tentukan sebuah citra digital karakter huruf '1' akan dianalisis bentuk menggunakan chain code. Tentukan faktor bentuk dari citra tsb.



Kode rantai = 06066644321022
 Total kode genot = 12
 11 11 ganjil = 2

• Menghitung variansi

$$D = 12 \times (1/2 \times 2.2)$$

$$= 14.828$$

• Menghitung luas area berdasarkan kode rantai:

Kode rantai	0	6	6	6	6	6	4	4	3	2
ordinal	6	6	5	4	3	2	1	1	1	2
area	6.0	0.0	0.0	0.0	0.0	0.0	-1.0	-1.0	-1.5	0.0

Kode rantai	1	0	2	2
ordinal	3	4	4	5
area	3.5	4.0	0.0	0.0

$$\begin{aligned} \Sigma \text{ area} &= 6.0 + 0.0 + 0.0 + 0.0 + 0.0 + 0.0 + (-1.0) + (-1.0) + (-1.5) + 0.0 + 3.5 + 4.0 + 0.0 + 0.0 \\ &= 10 \end{aligned}$$

• Menentukan Faktor (Dantur. / s)

$$s = \frac{(14.828)^2}{10}$$

$$= \frac{219.869584}{10} = 21.987$$